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EFFECTS OF INCREASED TEMPERATURE ON COLD-WATER ORGANISMS

George R. Snyder and Theodore H. Blahm

Man must resolve the problem of thermal pollution if he wishes to maintain the quality of his environment. The magnitude of this problem varies throughout the world as it does within the continental borders of the U. S. The effects of thermal pollution on aquatic ecosystems have been stated in extremes—from the destruction of most fauna to actual benefits.

Most experts agree on general points concerning thermal increases in the aquatic environment: (a) temperature will increase, (b) the major cause will be the expansion of the electrical power industry, (c) and the amount of waste heat ejected from thermal electric plants is predictable. The specific questions are where and how this heat will be dispersed and what effect this increase will impose on aquatic organisms.

The effects of temperature on fish are fairly consistent in the laboratory but inconsistent in nature. Fish succumb faster or slower than predicted through laboratory studies where variables are controlled. Uncontrolled releases of heat can destroy, dislodge, or debilitate portions of the aquatic biota, but controlled releases of heat can be utilized to benefit organisms. Consequently, each ecosystem must be studied through interdisciplinary research to determine

how to achieve maximum physical, chemical, and biological benefits.

The indigenous fish of major economic concern in the Pacific Northwest are primarily cold-water species; many of them are anadromous. The distribution of cold-water species is circumpolar. The problem is not restricted, therefore, to the Pacific Northwest. The family Salmonidae is also of major concern in the Great Lakes region and on the coasts of Asia, Europe, Canada, and New England.

The objective of this paper is to emphasize that increased water temperatures can produce both detrimental and beneficial effects and that cold-

TABLE I.—Installation Schedule for Thermal Power Plants*

Project	Capacity (Mw)	Scheduled Date of Operation	Ownership†
Centralia 1	700	September 1971	} Joint Ownership‡
Centralia 2	700	July 1973	
Nuclear 1	1,000	December 1974	Private
Nuclear 2	1,000	March 1976	Private
Nuclear 3	1,000	April 1977	Private
Nuclear 4	1,000	October 1978	Private
Nuclear 5	1,000	April 1980	Private
Nuclear 6	1,000	December 1981	Private
Nuclear 7	1,000	April 1982	Public
Nuclear 8	1,000	April 1983	Private
Nuclear 9	1,000	May 1984	Private
Nuclear 10	1,000	April 1985	Private
Nuclear 11	1,000	January 1986	Public
Nuclear 12	1,000	March 1986	Private
Nuclear 13	1,000	January 1987	Private
Nuclear 14	1,000	May 1987	Public
Nuclear 15	1,000	November 1987	Private
Nuclear 16	1,000	July 1988	Private
Nuclear 17	1,000	September 1988	Public
Nuclear 18	1,000	July 1989	Private
Nuclear 19	1,000	November 1989	Public
Nuclear 20	1,000	February 1990	Private

* From summary of preliminary report, "A Ten-Year Hydro-Thermal Power Program for the Pacific Northwest," Bonneville Power Admin., Portland, Ore., October 9, 1968.

† Ownership of each whole nuclear plant based on power requirement. Any or all plants could be jointly owned by private and public system.

‡ Centralia 1 and 2 ownership computed at 28 percent public agencies and 72 percent private utilities.

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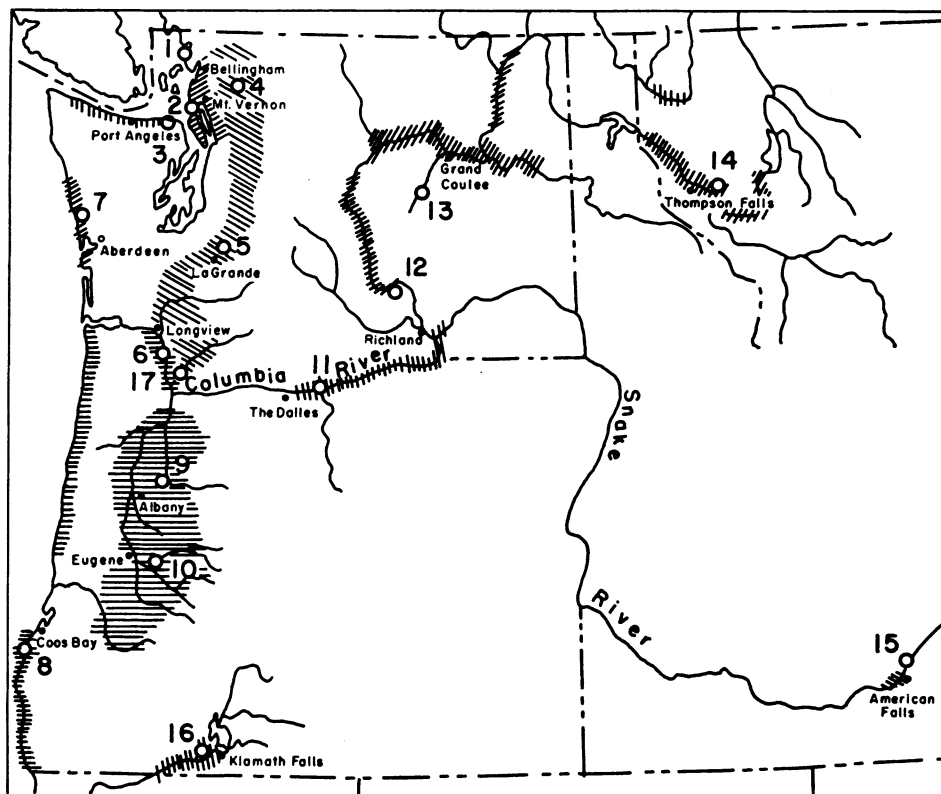


FIGURE 1.—Sites of proposed thermonuclear electric plants in the Pacific Northwest. Crosshatched areas show preferred areas for nuclear sites; circles indicate sites that were investigated.

water fish can succumb to direct and indirect effects of increased temperature.

Detrimental Effects

The effects of increased water temperature probably will be most detrimental in geographic areas where production of indigenous fish is marginal. Recently much controversy has been stimulated by the proposed installation of 20 thermonuclear electric plants (1,000 Mw/plant) in the Pacific Northwest by 1990 (Table I). Four of these plants, representing a production potential of 9,000 Mw will be on the Columbia River (Figure 1).

Summer water temperatures in the Columbia River are above the preference of important stocks of Pacific salmon (genus *Oncorhynchus*) and

steelhead trout (*Salmo gairdneri*). The controversy stimulated by the proposed thermonuclear plants does not question the need for power; the controversy rages over the dissipation (in air or water) of waste energy (heat).

The impact of waste heat from thermal plants on aquatic organisms is felt in three specific areas: (a) in the condensers where temperature increases sharply (Figure 2); (b) in thermal plumes issuing from discharge tubes where temperature differences between the plumes and river are sharp; and (c) throughout the entire receiving body of water after the complete diffusion of the thermal plume. Fish are subjected to sudden temperature increases from 8° to 17°C in the thermal plumes and to lesser increases in the entire river cross section.

The general impact of temperature increases on fish fauna of a biome may be described through a discussion of direct and indirect effects of heat.

Direct Effects

The direct effect of increased water temperature on a species of fish varies with the season, stage of life cycle, size, age, body condition and functions, and acclimation temperature.

The lethal effect of temperature on fish, described by some biologists as heat death, seems to be expressed in two different ways (Figure 3): (a) a rapid death (thermal shock) in seconds or minutes at high temperatures, and (b) a slower death in hours or weeks at moderate temperatures.

Information on the direct effects of heat on fish can be obtained from controlled laboratory experiments where

other conditions are held constant while temperature is varied.

Rapid Death from Heat:—Experimentation on sharp temperature increases (thermal shock) with anadromous fish was initiated by the Bureau of Commercial Fisheries, U. S. Department of the Interior, at the Seattle Biological Laboratory, Seattle, Wash., and on the Columbia River near Prescott, Ore.

Preliminary findings from the Prescott studies indicate that small (35- to 50-mm fork length) chinook salmon (*O. tshawytscha*) and chum salmon (*O. keta*) succumb rapidly (4 to 11 sec) to sharp temperature increases (Table II). The juvenile salmon died within seconds when subjected to water temperatures of 32°C. Thus, if ambient river temperature were 16°C, it would be necessary to raise it some

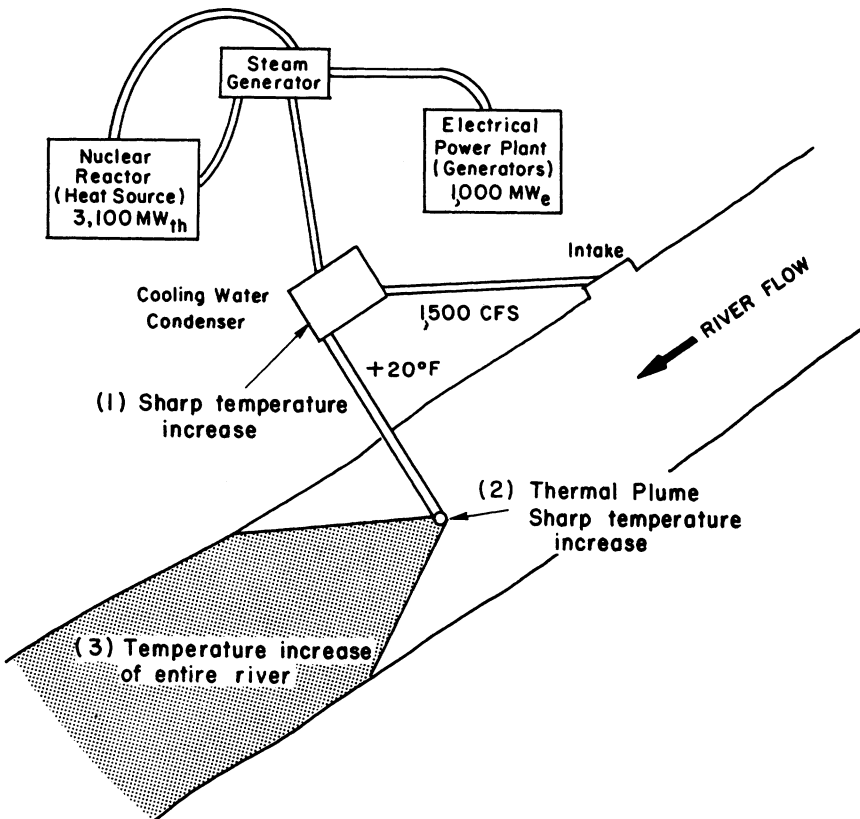


FIGURE 2.—Three areas near, or in, thermal plants where temperature increases will affect aquatic biota. [Cfs $\times 1.7$ = cu m/min, $0.555 (^{\circ}\text{F} - 32) = ^{\circ}\text{C}$.]

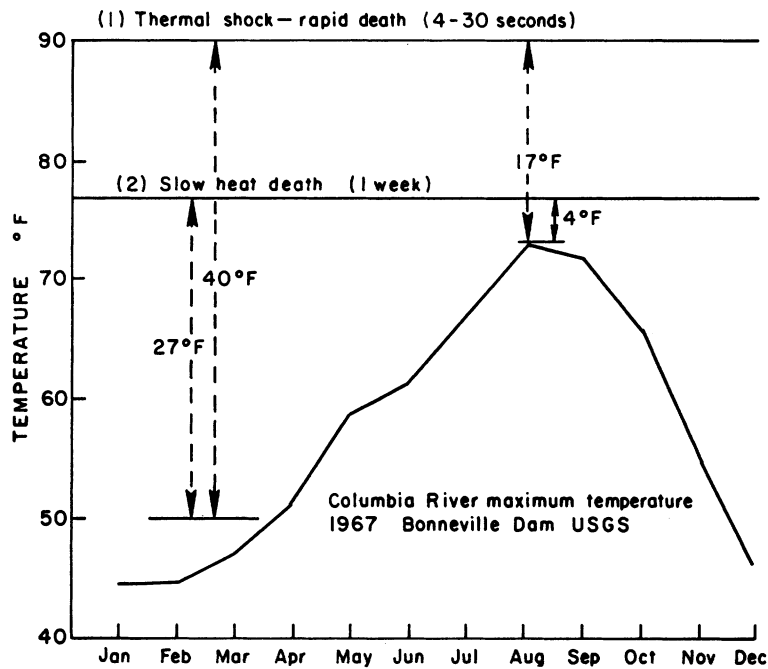


FIGURE 3.—Mortalities from temperature increases on salmon are caused by (a) rapid death from thermal shock, and (b) slow death from heat. [0.555 (°F – 32) = °C.]

16°C to produce rapid mortality. At river temperatures of 24°C, an increase of only 8°C would produce rapid mortality. An important aspect of these studies is that the 32°C water temperature, used as an upper limit in the

Prescott studies, closely approximates the thermal characteristics of the “N” reactor’s thermal plume at Hanford, Wash.¹

Biologists are concerned about the sharp increases in temperature, the characteristic of thermal plumes, and the stress of thermal shock on juvenile fish migrating downstream. It is possible that fry and fingerlings will die as they are swept into these plumes as a result of their apparent inability to detect and swim out of the zone of heated water.

Death of juvenile salmon from sharp increases in temperature is usually preceded by a loss of equilibrium. This phenomenon, described by Dean and Coutant,² may be a useful tool in the establishment of a sublethal measure of stress, but loss of equilibrium in a river could result in loss of the fish to a predator.

Blahm and McConnell³ subjected adult eulachon (*Thaleichthys pacificus*) from the Columbia River to a

TABLE II.—Mortality of Juvenile Chinook and Chum Salmon Subjected to Various Levels of Water Temperature*

Test Water Temperature (°C)	Increase (°C)	Time (sec)		
		First Mortality	50% Mortality	100% Mortality
Chinook Salmon†				
10 (Control)	+0	—	—	—
18.3	+8.3	—	—	—
21.1	+11.1	—	—	—
26.7	+16.6 (30°F)	100	240	2,220
32.2	+22.2 (40°F)	4	8	11
Chum Salmon‡				
15.6 (Control)	+0	—	—	—
21.1	+5.5	—	—	—
23.9	+8.3	—	—	—
26.7	+11.1	2,640	2,970	3,960
29.4	+13.8	15	26	60
32.2	+16.6 (30°F)	6	10	15
37.8	+22.2 (40°F)	4	5	7

* All fish were subjected to test temperature for 1 hr. Survivors were held post-test for 72 hr.
† Tested April 1968; fork lengths, 35 to 50 mm.
‡ Tested April and May 1968; fork lengths, 35 to 50 mm.

series of temperature increments (3°C) from 10°C (control) to 32°C. The smelt succumbed rapidly at high temperatures, but also were adversely affected (Table III) by increases as small as 8.3°C. At the end of the post-test holding period of 50 hr, all female smelt subjected to temperature increases failed to deposit their eggs. Most of the control fish that were kept at river temperature, however, had deposited their eggs in the holding tanks.

Slow Death from Heat:—Slow death from the direct effect of heat on Pacific salmon and steelhead trout (*Salmo gairdneri*) can be predicted, within certain limits, from laboratory experiments conducted by Brett.⁴ He reported that juvenile Pacific salmon and steelhead trout have a temperature tolerance zone between 4° and 20°C. Immediately outside the tolerance zone is a zone of resistance in which they can exist but only at the expense of such functions as growth, migration, feeding, and maturation. Beyond the zone of resistance are lethal zones that range from about 13° to 25°C at the upper levels, depending on acclimation temperatures. Brett⁵ stated that no species of juvenile salmon, including chinook, pink (*O. gorbuscha*), sockeye (*O. nerka*), chum, and coho (*O. kisutch*) could tolerate exposure of 1 wk to temperatures exceeding 25°C.

Some temperatures in the Pacific Northwest during the summer reach well into the upper resistance zone (and possibly into the lethal zone) for juvenile Pacific salmon and steelhead trout (Figure 4). Jones⁶ has pointed out, "In the warm season of the year, when temperatures rise to 22° to 24°C, a further rise of 1° or 2°C may be sufficient to result in heat death."

Although laboratory experiments on the general tolerance zones for salmon and trout have been extensive, little information is available on tolerance

TABLE III.—Mortality of Adult Eulachon Subjected to Various Water Temperatures*

Test Water Temperatures (°C)	Temperature Increase (°C)	Time (min)		
		First Mortality	50% Mortality	100% Mortality
10 (Control)	0	—	—	—
13	+3.3	—	—	—
15	+5.5	—	—	—
18	+8.3	36.0	19.2	—
21	+11.1	15.0	26.0	—
24	+13.8	5.0	14.0	51.0
27	+16.6	1.4	2.0	3.3
32	+22.2	0.3	0.3	0.3

* Tested March 1968. Ten fish used in each sample, survivors were held post-test for 50 hr.

levels under river conditions. For example, Dean and Coutant² found that Columbia River fall chinook salmon (juveniles) were less tolerant to temperature changes than chinook salmon tested in a British Columbia laboratory by Brett.

Only a limited amount of experimentation has been carried out on the direct effect of heat increases on adult salmon and trout. Nakatani⁷ reported that chinook salmon "jacks" acclimated to temperatures of 15.6°C died within 2 hr when subjected to 26.7°C water, whereas fish subjected to 23.9°C lived.

Indirect Effects

The indirect effects of water temperature on fish often pose the most important problems to fishery resource managers. Increasing water temperatures can act alone, or with other physical, chemical, and biological variables to produce less desirable, sometimes lethal, conditions for some fish. Increased water temperatures can delay migration of anadromous fish in rivers, induce stress, contribute to disease problems, and favor some species of fish that compete with (and prey on) important stocks of salmon and trout.

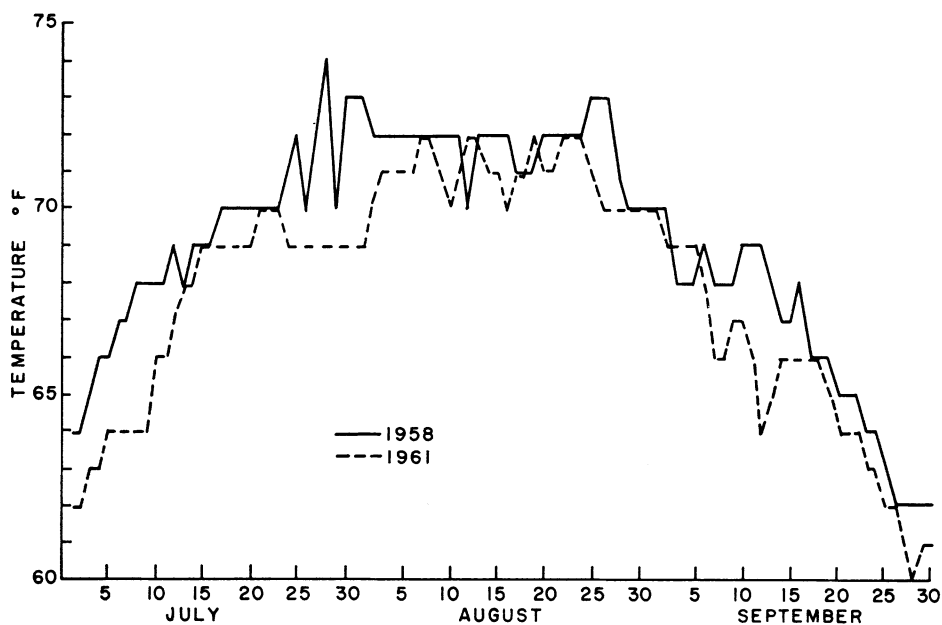


FIGURE 4.—Temperatures of the Columbia River at Bonneville Dam, 1958 and 1961. [0.555 (°F - 32) = °C.]

Probably the most publicized indirect effect of heat on salmon in the Pacific Northwest is the delay of upstream migration from water temperature blocks. These blocks can be either natural or man-made. Temperature blocks have been reported in the Okan-

ogan River⁸ and near the confluence of the Snake and Columbia rivers.⁹ The delay into the Snake River was caused by a temperature difference of 3.9°C between the Snake and Columbia rivers (Figure 5). The migration resumed when the difference ap-

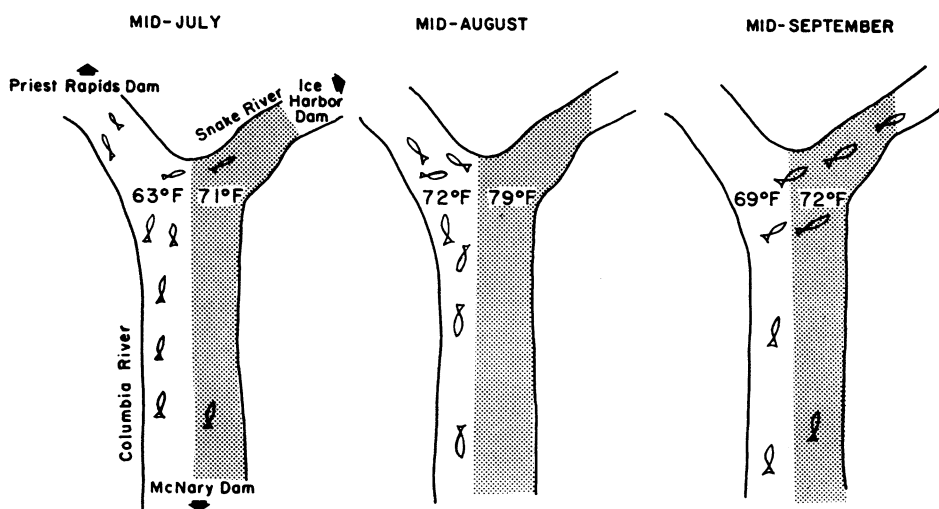


FIGURE 5.—Thermal conditions at the mouth of the Snake River in 1967 that blocked the upstream migration of adult chinook salmon and steelhead trout. [0.555 (°F - 32) = °C.]

proached 1.6°C. A delay in upstream migration of only 5 days caused significant mortalities in sockeye salmon of the Fraser River, B. C.; few of the salmon reached the spawning grounds when subjected to delays of 10 to 12 days.¹⁰

Nitrogen gas is introduced into the Columbia River at spillways of dams. The gas is held in solution in a supersaturated condition by the pressure of the deepened forebays. Gas bubble disease causes mortality in salmon and trout.¹¹ As water temperature is increased, it is possible to increase nitrogen saturation from a sublethal to a lethal level. Research on the interaction of increased temperature and nitrogen gas saturation on the mortality of chinook salmon "jacks" has shown that death occurs within 24 hr at 134 percent nitrogen saturation and 22°C.¹² Both conditions could occur naturally in the Columbia River. Although the time and occurrence of N₂ gas saturations in the Columbia River have been described adequately,¹¹ little is known about the total effect of nitrogen supersaturation on the stocks of anadromous fish.

Lack of oxygen can affect all life stages of finfish and shellfish. Temperature increases reduce the availability of dissolved oxygen (DO) in water, yet the same increases require that fish utilize more oxygen for body maintenance. The effect of increases of water temperature on oxygen saturation and the effect of these changes on fish have been extensively covered.^{6, 13}

Adams¹⁴ reviewed the literature concerning thermal electric plant discharges into freshwater ecosystems. Twenty thermal electric plant sites were included. Fish were studied at 12 sites, invertebrates at 10, and plankton at 4. Adams concluded that heated discharges decreased the cold-water species and increased the warm-water species. Thus, if the ecosystem of an entire river is subjected to a gradual

change in temperature, species composition could be affected. Novotny¹⁵ clarified this relation between cold-water species and warm-water predators and competitors in the Snake River (Figure 6). Increased water temperatures favor some species of predators and competitors at the expense of cold-water species.

Tarzwel¹⁶ explained this relation, "Increased temperature does not necessarily mean cold water fish will be replaced by warm water fish by mortality of the cold water fish. Often the warm water fish will increase in number due to their better adaptability to the warm water, and the cold water fish will decrease due to replacement."

Mortality of juvenile sockeye salmon infected with low-virulence strains of *Chondrococcus columnaris* increases as water temperatures increase.¹⁷ Mortalities are substantial before water temperatures reach 18°C. Ordal and Pacha¹⁸ have stated that salmon and trout in the Columbia River are infected by *C. columnaris* and that slight increases in river temperature could accelerate the effect. *Ceratomyxa*, a recently isolated disease-producing organism that infects salmon and trout, is associated with water from impoundments. The disease seems to be most virulent when water temperatures exceed 16°C. These diseases, *C. columnaris* and *Ceratomyxa*, affect salmon and trout that migrate upstream during the summer and early fall when temperatures exceed 16°C.

An examination by Mihursky and Kennedy¹⁹ of lethal temperatures of some important planktonic forms in the Patuxent River estuary, has shown that many species have narrow temperature tolerances. Similar planktonic fauna are utilized in estuaries of the Pacific Northwest as food by juvenile salmon and trout. Sims²⁰ observed that some juvenile salmon stay in the estuary of the Columbia River as long as 8 wk before they migrate to sea. If the present planktonic fauna

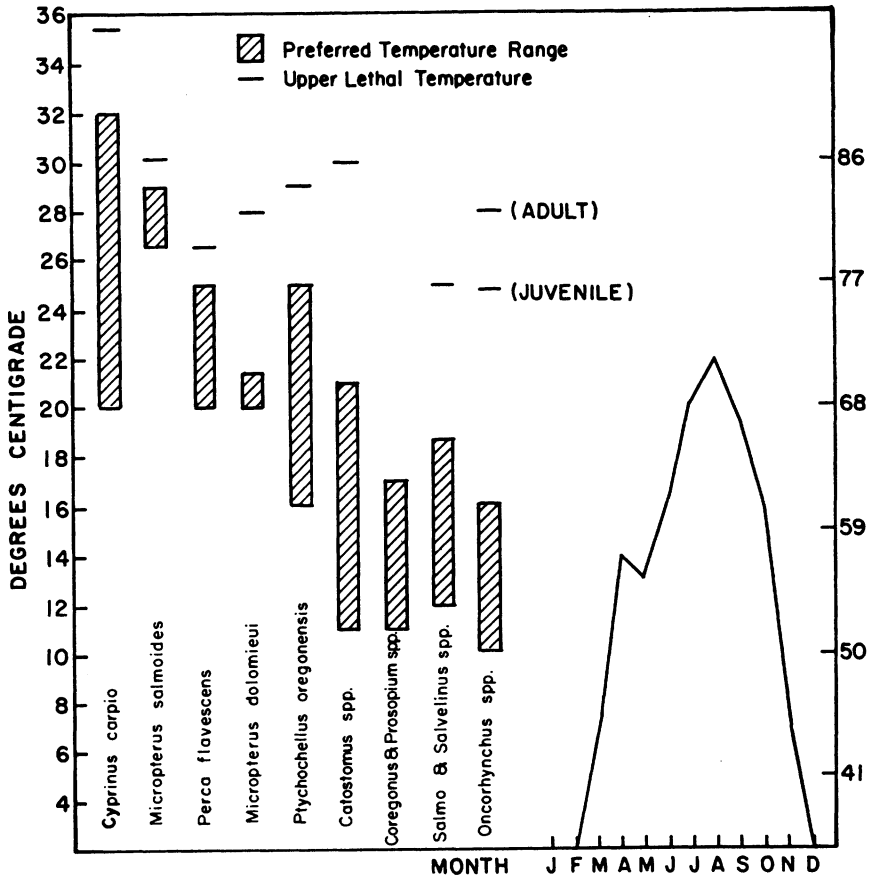


FIGURE 6.—Preferred and lethal temperatures of some Columbia Basin fish compared with water temperatures of the lower Snake River. (Note: Scale on right side of figure shows °F.)

are reduced or eliminated from the Columbia River estuary by elevated temperatures, the downstream migrants will have to feed on different organisms. There is no way to predict how a “forced” change in diet will affect important stocks of salmon and trout in the Columbia River.²¹

Beneficial Effects

Thermal discharges of power plants seem to have some beneficial uses; increased water temperatures can increase production of finfish and shellfish if properly controlled. A discussion of this potential was given by Gaucher,²² who described increases in production of freshwater and saltwater

organisms cultivated under controlled conditions.

There are many examples of increases in catch of sport fish near the cooling-water discharge area of thermal plants; the improved fishing usually is documented in temperate zones with warm-water and nonmigratory species of fish. Some question exists about the beneficial aspects of these outfall areas; although success in fishing apparently increases during the winter, it is nonexistent during the summer. Trembley²³ reported that indigenous species are “attracted to and invade” the heated water discharge from a power station on the Delaware River from late September until early June. The 40 species of fish at the site, however,

were primarily forage and warm-water fish. Elser²⁴ documented angling returns from natural and artificially heated water near an electric generating station on the Potomac River in Maryland. He stated that substantially more fish were landed per fishing trip except during July and August when water temperatures approached 32°C. Again the principal species were warm-water and forage fish.

The waste heat from thermal plants in the USSR has extended production of carp to areas formerly too cold to support this species. Beneficial effects from waste heat seem more likely in the temperate zone when the heat is applied to marine environments. Sole and oyster production has been increased in Great Britain through the controlled use of waste heat.²⁵ Sole now attain market size in 1 yr where formerly 3 yr were required.

Naylor²⁶ suggested that discharge of waste heat into saltwater should be carried out on open coastlines near the intertidal zones, not in deep water. He supported this view with data that point out the narrow temperature tolerance of pelagic and deepwater species and wide temperature tolerance of intertidal organisms.

Oyster production in Puget Sound is limited by water temperatures often far below desirable limits for spawning. Thus, seed oysters must be imported in large quantities to augment local supply. Production in the Pacific Northwest could probably benefit from selected locations through controlled discharges of waste heat and from raft and rack culture techniques. Research is needed to determine levels of production at increased temperatures. If the effects are to be beneficial in all stages of the life cycle, one should know also if increased temperatures are detrimental to other important species.

Raceway cultivation for freshwater and saltwater species of finfish and

shellfish was recommended by Gauthier.²² It would seem from his paper that the controlled addition of waste heat to freshwater environments of the upper temperate and subarctic zones would also prove to be beneficial for raceway cultivation of some species of Pacific salmon and trout.

Conclusions

Ejection of heat from large thermoelectric plants into the aquatic environment could result in declines of cold-water fish. It is apparent that some aquatic organisms indigenous to the Pacific Northwest have low tolerance to increases of temperature. Research is needed to define clearly the thermal tolerance and survival of these organisms over a wide range of environmental conditions.

Perhaps benefits could be realized in marine aquaculture if economically important species could be reared in an environment supplied with heated water discharged from large thermoelectric complexes. Obviously, research is needed to study the beneficial effects of heat on the marine environment and to determine what organisms would be affected by thermal changes. In the interim, thermal electric plants should not be placed on waters with temperatures that approach the upper limits of tolerance of the fish important to sport and commercial fisheries unless adequate facilities for offstream cooling are provided.

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